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SUPERSEDING
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MILITARY SPECIFICATION

ROCKET, HE, 66MM, ANTITANK, M72A2 W/COUPLER LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally requested in the drawings and provides the quality assurance and acceptance provisions for the loading, assembling, and packing for one type of antitank weapon designated as Rocket, HE, 66mm, Antitank: M72A2 W/Coupler.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or requests for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons; General Specification for
MIL-S-12560	Steel, Armor Plate, Combat Vehicle Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1340

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DRAWINGS

ARMY

9210200	Launcher, Rocket, 66mm: M72A1
9220248	Warhead, 66mm, Rocket Assy, HE: M18A1
9220251	Closure
9223224	Fuze, Rocket: M412A1
9227925	Box, Fiberboard, Packing, Ammunition for Rocket, HE, 66mm, Antitank: M72A1 or M72A2 W/Coupler
9227926	Box, Ammo Packing Wood Wirebound for Rocket, HE, 66mm, Antitank: M72A1 or M72A2 W/Coupler
9235663	Rocket, High Explosive, 66mm, M72A1
9244054	Rocket, HE, 66mm, Antitank: M72A2 W/Coupler
9244140	Ring, Seal
9246403	Calibration Standard
9251970	Socket, Connector
9251978	Cap, Base Connector
9251980	Flexible Cable Assembly
9270271	Interface Ring
9271064	Tape, Impregnated
9271631	Rocket, High Explosive, 66mm, M72A1 W/Coupler
10048508	Charge Assembly (Propellant)
10048509	Rocket Motor: M54
10048528	Nose Cap Assembly
10048662	Box, Fiberboard, Ammunition for Integral Igniter, Motor Rocket: M56
10048775	Integral Ignitor Motor Rocket: M56 Loading Assembly

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PUBLICATIONS

ET-9244054 Equipment Tabulation

3. REQUIREMENTS

3.1 Material. Materials and components utilized in the loading, assembling, and packing of the weapon system shall comply with drawings 9244054, 9227925, 9227926, their subdrawings, and specifications listed thereon.

3.2 Complete weapon. The weapon shall be in strict accordance with drawing 9244054, its subdrawings, and specifications listed thereon.

3.3 Propellant charge. The propellant grains shall not be broken, cracked, or damaged, and the charge assembly shall be in strict accordance with drawing 10048508, its subdrawings, and specifications listed thereon.

3.4 Fuze. The fuze shall be in strict accordance with drawing 9223224, its subdrawings, and specifications listed thereon. The fuze shall be in the unarmed condition when assembled to the warhead.

3.5 Rocket motor metal parts assembly.

3.5.1 Motor case and nozzle assembly. The motor case and nozzle assembly shall be in strict accordance with drawing 10048509, its subdrawings, and specifications listed thereon. The nozzle exit cone area of the motor body shall be free from all external and internal cracks and metallic discontinuities when examined by "eddy current" equipment adjusted to a sensitivity to disclose metallic discontinuities of 0.062 inch maximum diameter hole. The rocket motor body shall be free from damage including but not limited to nicks, scratches, gouges, or any visible breaks in the protective coating that exceed 0.002 inch in depth, apparent cracks, or material bruises.

3.5.2 Closure. The closure shall be in strict accordance with drawing 9220251, its subdrawings, and specifications listed thereon. The closure shall be free from cracks, as defined by the standards of drawing 9246403, when tested with ultrasonic equipment. Before application of the coupler, the closure shall be free from damage including, but not limited to, nicks, scratches, gouges, or any visible breaks in the protective coating that exceed 0.005 inch in depth, or apparent cracks when examined under three-power magnification.

3.5.3 Physical strength of exit cone. The motor body shall not crack or undergo any permanent deformations when stressed within the following limits at the locations specified as a part of the test specified in 4.4.2. Stress readings shall indicate the hoop stress to which the motor exit

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cone has been subjected. Maximum stress shall not exceed 70,000 pounds per square inch. Minimum stress at all unspecified exit cone locations shall be that resulting from the application of the specified test equipment to achieve the minimum stress levels specified at the specified locations. Location of stress measurement points shall be measured along the longitudinal axis of the motor body beginning at the lip of the exit cone. Stress measurement points and minimum stress levels shall be:

3/16 inch	-	21,000 pounds per square inch
1 inch	-	10,000 pounds per square inch
1 3/8 inch	-	34,000 pounds per square inch

At and forward of the aftermost 0.982 inch nozzle inside diameter location through the remainder of the motor body stress shall be that resulting from 10,000 pounds per square inch minimum to 10,800 pounds per square inch maximum internal pressure.

3.6 Warhead. The warhead assembly shall be in strict accordance with drawing 9220248, its subdrawings, and specifications listed thereon.

3.6.1 Continuity of cable assembly in loaded warhead without fuze. Before assembling fuze to the loaded warhead, the resistance of each conductor circuit shall not exceed 0.5 ohms.

3.6.2 Electrical resistance. The warhead assembly (drawing 9220248) consisting of assembled warhead, fuze and closure (less nose cap assembly, drawing 10048528) shall exhibit an electrical resistance of not less than one megohm when measured between the positive and negative contacts of the nose cap connector.

3.7 Igniter assembly. The igniter assembly shall be in strict accordance with drawing 10048775, its subdrawings, and specifications listed thereon. Before assembly of the igniter into motor bodies, the moisture content of the black powder charge shall not exceed 0.50 percent.

3.8 Seal between igniter and motor body. The seal between the igniter and motor body shall be air tight when subjected to an air pressure differential of 10 psi, minimum.

3.9 Cure strength of unassembled coupler. The unassembled coupler, a 14-inch length of impregnated tape (drawing 9271064), covering a detail of drawing 9271631 shall not deflect under its own weight more than two inches when cantilevered flat and horizontally four inches plus or minus one-fourth inch, after being cured at 85 degrees to 125 degrees Fahrenheit for sixteen hours.

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3.10 Strength of warhead-closure joint. The warhead-closure joint shall withstand a pull of 200 plus or minus ten pounds minimum without movement or separation.

3.11 Strength of motor-closure assembly. The assembled rocket (drawing 9235663) after cure shall withstand a disassembly torque of 100 inch pounds without movement occuring between the rocket motor and the closure.

3.12 Rocket weight. The weight of the assembled rocket with igniter shall be as specified in drawing 9271631.

3.13 Launcher. The launcher shall be in strict accordance with drawing 9210200, its subdrawings, and specifications listed thereon.

3.14 Performance.

3.14.1 Weapon functioning. The weapon shall function on the first attempt, and shall not have an ignition delay in excess of one second from the activation of the trigger mechanism to the emergence of the rocket from the muzzle of the launcher. The fuze or warhead shall not function prematurely. The motor or closure shall not fracture or separate. The launcher shall not explode or implode.

3.14.2 Warhead functioning. The warhead shall function electrically upon impact with a target of any angle from 30 degrees through 90 degrees to the line of flight of the rocket at the target. The warhead shall penetrate five inches of MIL-S-12560, Class 1, armor plate placed at an angle of 30 degrees to the line of flight of the rocket and centered a minimum of 100 feet from the muzzle end of the launcher. The warhead shall function on graze impact with hard packed earth at an angle six to ten degrees with the line of flight at a minimum of 200 meters range.

3.14.3 Accuracy. The rocket shall have a first round hit probability of 94 percent against a vertical 7-1/2 x 7-1/2 foot target at seventy degrees Fahrenheit when stand fired at 200 meters range.

3.15 Environmental and safety.

3.15.1 Drop impact. The weapon shall not detonate or become propulsive and shall be safe to fire after being dropped once in its carrying condition from a height of five feet onto concrete.

3.15.2 Immersion. The weapon shall function and perform in accordance with 3.14.1 and 3.14.2 after being submerged in the closed launcher under three feet of water for two hours.

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3.15.3 Low temperature. The weapon shall function and perform in accordance with 3.14.1 and 3.14.2 when conditioned and fired at -40 degrees Fahrenheit.

3.15.4 High temperature. The weapon shall function in accordance with 3.14.1 and 3.14.2 when conditioned to and fired at +140 degrees Fahrenheit.

3.15.5 Transportation vibration. The weapon shall meet the requirements of 3.14.1 and 3.14.2 after being subjected to vibration in accordance with MIL-STD-810, Method 514, Procedure X, Curve AA.

3.16 Workmanship. All parts, subassemblies, and assemblies shall be free of chips, dirt, grease, rust, and other foreign material that may impair proper functioning or deterioration in storage. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent used. Painted or coated surfaces shall be restored when these surfaces have been marred or scratched. All marking shall be clean and sharply defined.

3.17 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the specified quality assurance provisions and contract requirements.

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b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed in this section is identified by Equipment Tabulation ET-9244054 (see 6.3).

4.1.3 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction survey and whenever inspection equipment drawings are needed.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection, examination and test (see 4.3).

b. Quality conformance inspection, examination and test (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample to a Government approved facility as designated by the contracting officer (see 6.6) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 150 complete weapons and 10 each of every assembly, subassembly and component which have been produced by the contractor. First article samples of the major components and subassemblies of the M72A2 weapon shall be submitted to the Government

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in accordance with their respective contracts and specifications. For information purposes, these are listed below.

Launcher	MIL-L-60385
Sight	MIL-S-14947
Fuze	MIL-F-50415
Fuze	MIL-F-50407
Warhead	MIL-W-50275
Warhead	MIL-W-50422
Nose Cap	MIL-C-60006
Igniter	MIL-I-60008
Motor	MIL-M-60003
Closure	MIL-C-50402
Propellant Charge	MIL-P-60009

Submission of the first article sample under this specification shall be accomplished utilizing components, assemblies and subassemblies form qualified vendors having passed first article tests required by above specifications and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Any time there is a change in interfix lot number in closure or motor body assemblies or propellant charge assemblies, one first article sample must satisfactorily complete the accuracy test portion of first article sample testing prior to the utilization of the changed interfix lot numbered parts in production assembly operations. Prior to submission of the first article sample, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance of materials. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.4.2 and 4.4.3 of this specification and any or all requirements of the applicable drawings. The proving ground test samples submitted as first article production shall be tested in accordance with 4.3.4.

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4.3.3 Rejection. If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspections and/or testing upon any failure of an assembly or component to comply with any of the stated requirements. With the exception of accuracy, the rejection criteria for the proving ground portion of the first article sample shall be in accordance with Table I. The first article sample shall also be rejected if,during the 70°F accuracy phase, the displacement of the average center of impacts from the center of the target, plus or minus the product of 1.55 and sample standard deviation exceeds 45 inches vertical distance from target center. Sample standard deviation shall be computed using the formula in 6.4. For defect code 17013, one partial penetration shall be disregarded providing that bulging or cracking at the back side of the plate occurs.

4.3.4 Proving ground test. The first article sample of 150 complete weapons shall be tested at an approved Government proving ground. Quantities allocated to the respective phases of test are indicated on Table I. Graze functioning and penetration tests shall be conducted prior to -40°F flight tests. Any misfire during graze functioning or penetration tests shall count against the total misfire criteria but shall not count as a warhead or fuze failure. Any fuze failure during penetration tests shall count against the total fuze failure criteria but shall not count as a penetration failure. Weapons which misfire during graze tests or during penetration tests and rounds suffering fuze failure during penetration tests shall be replaced with samples from the -40°F flight tests until the total required quantity of these specific functions have been tested. Low temperature tests (-40°F) shall be conducted using balance of samples available. For the test to determine compliance with acceptance criteria for 70°F accuracy firings, $\bar{x} \pm 1.55S = < 45$ inches (see 6.4) applied to the vertical distance shall be used. The complete Proving Ground test shall be performed regardless of any early rejection indications, except for occurance of critical defects. Defects, classifications, and code numbers for the proving ground portion of the first article sample are listed in 4.4.4.2.

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TABLE I

FIRST ARTICLE SAMPLE
ACCEPTANCE/REJECTION CRITERIA PROVING GROUND TESTS

DEFECT CODES	QUANTITY IN TEST / TEST / DEFECT	10 TV	10 DROP	10 -40°F GRAZE	20 -40°F FUNCTION	20 +140°F FUNCTION	20 WATERPROOF PENETRATION	60 70°F ACCURACY	150 QUANTITY CONSIDERED	CRITERIA ACC.	CRITERIA REJ.
17001	Premature	X	X	X	X	X	X	X	150	0	1
17002	Motor Rupture	X	X	X	X	X	X	X	150	0	1
17003	Closure Rupture	X	X	X	X	X	X	X	150	0	1
17004	T_I > 3 Seconds	X	X	X	X	X	X	X	150	0	1
17005	Launcher Fails	X	X	X	X	X	X	X	150	0	1
17006	Warhead Elec Failure	X	X		X	X	X	X	140	5	6
17007	Accuracy							X	60	*(1)	*(1)
17008	Misfire	X	X	X	X	X	X	X	150	4	5
17009	**T_L 1 to 3 Seconds	X	X	X	X	X	X	X	150	1	2
17010	***T_I 1 to 3 Seconds	X	X	X	X	X	X	X	150	1	2
17011	Parts Missing or damaged	X	X	X	X	X	X	X	150	0	1
17012	Target Miss	X	X		X	X		X	120	4	5
17013	Penetration Failure						X		20	1	2
17014	Fuze Armed	X	X						20	0	1
17015	Warhead Dud	X	X		X	X	X	X	140	1	2
17016	Fuze Fails to Function			X					10	1	2
	Test Method	4.5.2.2.1	4.5.2.2.2 / 4.5.2.2.3		4.5.2.1.3	4.5.2.1.3	4.5.2.1.1	4.5.2.1.2			
	Requirement	3.15.5	3.15.1	3.14.2	3.15.3	3.15.4	3.15.2	3.14.3			

*(1) See 4.3.3

**T_L = Time to Launch (see 6.7)

***T_I = Time to Ignition (see 6.8)

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4.4 Quality conformance, inspection, examination and test.

4.4.1 Lot formation. The term "inspection lot", as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Propellant assemblies from not more than one lot from one manufacturer.

b. Parts and parts assemblies from not more than one interfix lot number from one manufacturer.

(Applicable to each part or parts assembly individually).

c. Loaded warheads from not more than five (5) lots.

d. Closures from not more than two (2) lots from one manufacturer.

e. Motors from not more than two (2) lots from one manufacturer.

f. Fuzes from not more than one lot from one manufacturer.

4.4.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.4.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. When a 100% inspection is specified, all defective components shall be removed from the lot. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous

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sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph of this section, except when 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.1.2.

4.4.2.1 Body loading assembly (see dwg. 9220247 covering a detail of dwg. 9220248).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Assembly, including explosive, dented, distorted, or otherwise damaged	Visual	01001
102.	Evidence of foreign material	Visual	01002
Minor:	None defined.		

4.4.2.2 Closure (see dwg. 9220251 covering a detail of dwg. 9220248).

Categories	Defects	Method of Inspection	Code No.
Critical:	100% Inspection		
1.	One or more test stamps missing from closure (hydrostatic or ultrasonic)	Visual	02001
Major:	AQL 100%		
101.	Evidence of thread damage, crossed or burred	Visual	02002
Minor:	None defined.		

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4.4.2.3 Warhead, 66mm, rocket, HE: M18A1 (see dwg. 9220248 covering a detail of dwg. 9235663, prior to bonding of closure).

Categories	Defects	Method of Inspection	Code No.
Critical:	100% Inspection		
1.	Fuze in armed position	Visual	03001
Major:	AQL 0.65 Percent		
101.	Seal ring (9244140) missing	Visual	03002
102.	Interface ring (9270271) missing	Visual	03003
103.	Flexible cable assembly not connected to fuze pins	Visual	03004
104.	Base connector cap (9251978) missing	Visual	03005
105.	Tape missing or improperly applied	Visual	03006
Minor:	None defined.		

4.4.2.4 Warhead, 66mm, rocket, HE: M18A1 (see dwg. 9220248 covering a detail of dwg. 9235663).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Nose cap assembly to flexible cable connection insecure	Manual	04001
102.	Gap between closure and interface ring and/or gap between interface ring and warhead	Visual	04002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.16)	Visual	04003

4.4.2.5 Charge assembly (see 3.3 and dwg. 10048508 covering a detail of dwg. 9235663).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Grain or grains split, cracked, broken or missing	Visual	05001
Minor:	None defined.		

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4.4.2.6 Igniter assembly (see dwg. 10048775 covering a detail of dwg. 9235663).

Categories	Defects	Method of Inspection	Code No.
Critical:	100% Inspection		
1.	Lid or parts loose or damaged to the extent that the explosive charge may leak out ----------	Visual	06001
Major:	AQL 0.40 percent		
101.	Transmission tube kinked or shows evidence of having been kinked ----------	Visual	06002
Minor:	None defined.		

4.4.2.7 Motor, rocket (see dwg. 10048509 covering a detail of dwg. 9235663).

Categories	Defects	Method of Inspection	Code No.
Critical:	100 % Inspection		
1.	One or more test stamps missing (Hydro, Force Cone) ----------	Visual	07001
Major:	AQL 100%		
101.	Evidence of thread damage, crossed or burred -----	Visual	07002
Minor:	None defined.		

4.4.2.8 Rocket, high explosive, 66mm, M72A1 (see dwg. 9235663, covering a detail of dwg. 9271631).

Categories	Defects	Method of Inspection	Code No
Critical:	100% Inspection		
1.	Damaged or missing coating on closure ----------	Visual	08001
2.	Failure of closure to meet visual examination requirement (see 3.5.2) ----------	Visual	08002
Major:	AQL 0.40 Percent		
101.	Paint removal improper ----------	Visual	08003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.16) ----------	Visual	08004

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4.4.2.9 Fiberglass tape (see dwg. 9270931).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Weight	Gage	09001
102.	Width	Gage	09002
103.	Contour	Gage	09003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.16)	Visual	09004

4.4.2.10 Rocket, high explosive, 66mm,M72A1 w/Coupler (see dwg. 9271631 covering a detail of dwg. 9244054).

Categories	Defects	Method of Inspection	Code No.
Critical:	100% Inspection		
1.	One or more test stamps missing from motor (loading plant force cone, eddy current)	Visual	10001
2.	Failure of motor to meet visual examination requirement (see 3.5.1). The cardboard retainer shall be positioned so as to enable the inspector to view the entire exit cone external surface area and the entire exit cone external and internal surface areas shall be inspected	Visual	10002
3.	Damage to motor closure or warhead following pull test	Visual	10003
Major:	100 percent inspection		
101.	Epoxy on warhead major diameter	Visual	10004
102.	Closure not concentric with body and rear of motor-	Gage	10005
Major:	AQL 1.0 Percent		
103.	Improper fin actuation	Visual-Manual	10006
104.	Loose or improperly positioned vinyl tape	Visual	10007
105.	Assembly incomplete or incorrect	Visual	10008
106.	Fins bent or damaged	Visual	10009

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Categories	Defects	Method of Inspection	Code No.
Major Cont'd:			
107.	Exposed fiberglass tape	Visual	10010
108.	Nose Cap not secure	Visual-Manual	10011
109.	Nose cap improperly aligned	Visual	10012
110.	Loaded fuze lot number missing or unidentifiable	Visual	10013
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.16)	Visual	10014

4.4.2.11 Rocket HE, 66mm, Anti-Tank: M72A2 w/coupler (see dwg. 9244054).

Categories	Defects	Method of Inspection	Code No.
Critical:	100 % inspection		
1.	Improper twist of round lock	Visual	11001
2.	Gouging or bending of motor exit cone resulting from twisting of round lock	Visual	11002
Major:	AQL 100%		
101.	Assembly force excessive	Gage	11003
Major:	AQL 0.65 percent		
102.	Components missing or improperly assembled	Visual	11004
103.	Firing pin housing not securely attached	Visual-Manual	11005
104.	Marking missing, misleading, or unidentifiable	Visual	11006
105.	Limited light sight marking incorrect or illegible	Visual	11007
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.16)	Visual	11008

4.4.2.12 Carton, prior to sealing (see dwg. 9227925 covering a detail of dwg. 9227926).

Categories	Defects	Method of Inspection	Code No.
Critical:	100% Inspection		
1.	Rounds reversed in position from carton nose end designation	Visual	12001
2.	Rear saddle missing or damaged	Visual	12002

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.40 percent		
101.	Component missing, damaged, or incomplete	Visual	12003
Minor:			
201.	Component packed incorrectly	Visual	12004
202.	Evidence of poor workmanship (see 3.16)	Visual	12005

4.4.2.13 Carton, Sealed (see dwg. 9227925 covering a detail of dwg. 10048675.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Carton damaged to the extent of exposing components	Visual	13001
102.	Barrier bag damaged to the extent of exposing carton	Visual	13002
Minor:	AQL 1.50 percent		
201.	Marking missing, misleading or unidentifiable	Visual	13003
202.	Handle missing or not properly attached	Visual	13004
203.	Center band strap on sealing strip missing or not properly attached	Visual	13005
204.	Sealing strips missing	Visual	13006

4.4.2.14 Box, packing, prior to closing (see dwg. 9227926).

Categories	Defects	Method of Inspection	Code No.
Critical:	100% Inspection		
1.	Combined cushion thickness insufficient on end opposite nose end of each carton	Visual	14001
2.	Carton reversed from nose end designation	Visual	14002
Major:	AQL 0.40 percent		
101.	Carton missing	Visual	14003
102.	Carton improperly packed in box	Visual	14004

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.00 percent		
201.	Packing material missing or not sufficient-	Visual	14005
202.	Evidence of poor workmanship (see 3.16)----	Visual	14006

4.4.2.15 Box, packing, closed (see dwg. 9227926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Board broken or split ----------	Visual	15001
102.	Car seal missing ----------	Visual	15002
103.	Binding wire missing or loose ----------	Visual	15003
Minor:	AQL 1.50 percent		
201.	Marking missing, misleading, or unidentifiable ----------	Visual	15004
202.	Handle missing ----------	Visual	15005
203.	Evidence of poor workmanship (see 3.16)--	Visual	15006

4.4.3 Testing (loading plant).

4.4.3.1 Physical strength of exit cone (see 3.5.3) - Critical defect - Code No. 07003. This test shall be performed 100 percent. The test shall be performed as specified in 4.5.1.1 using test equipment specified in 4.1.2. Defective motor bodies shall be rejected and removed from the lot.

4.4.3.2 Examination for physical defects in exit cone of motor body (see 3.5.1) - Critical defect - code No. 07004. This test shall be performed 100 percent. The test shall be performed as specified in 4.5.1.2 using test equipment specified in 4.1.2. Defective motor assemblies shall be rejected and removed from the lot. The test equipment shall be functionally verified by the Government at half hour intervals of production, and prior to and immediately following acceptance operations at the beginning and end of the work day. This verification shall consist of processing the standards specified in 4.1.2 and observing for the correct "accept" or "reject" signal and corresponding acceptance stamp appearing on acceptable bodies only. Malfunctioning test equipment shall be adjusted in the presence of the Government inspector. All test equipment verification for the Government shall be witnessed by the Government inspector who may also conduct verification at any unscheduled period or periods of production. The Government Inspector shall reject all motor bodies accepted since the last successful Government verification, subject to 100 percent reinspection by the contractor, if during any Government verification check the equipment fails to be in adjustment as determined by the test equipment standards.

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4.4.3.3 Test for cracks in the closure (see 3.5.2) - critical defect - Code No. 02003. This test shall be performed 100 percent as specified in 4.5.1.3 using test equipment specified in 4.1.2. Defective closures shall be removed from the lot.

4.4.3.4 Continuity of cable assembly in loaded warhead without fuze (see 3.6.1) - major defect - code No. 03007. The test shall be performed 100 percent as specified in 4.5.1.4 using test equipment specified in 4.1.2. Defective assemblies shall be rejected and removed from the lot.

4.4.3.5 Electrical resistance check of warhead assembly less nose cap (see 3.6.2) - major defect - code no. 04004. This test shall be performed 100 percent. The test shall be performed as specified in 4.5.1.5. using test equipment specified in 4.1.2. Defective assemblies shall be rejected and removed from the lot.

4.4.3.6 Closure/warhead joint pull test (see 3.10 and dwg. 9220248 critical defect - code no. 10015. This test shall be performed 100% as specified in 4.5.1.6 using test equipment specified in 4.1.2. Defective assemblies shall be removed from the lot.

4.4.3.7 Cure strength of unassembled coupler (see 3.9) - major defect - code no 16001. A sample of five (5) for each batch of mixed resin, fourteen (14) inch lengths of impregnated fiberglass tape, after cure cycle, shall be subjected to the test specified in 4.5.1.7 using test equipment specified in 4.1.2. The quantity of wrapped rounds represented by the batch of mixed resin shall be rejected if any length of tape fails the test.

4.4.3.8 Moisture content of black powder charge in the igniter body (see 3.7) - major defect - code no. 06003. A sample of 1 igniter shall be selected from each box (10048662) of igniters and tested as specified in 4.5.1.8. The lot shall be rejected if the moisture content of one or more charges exceeds the requirement specified.

4.4.3.9 Seal between igniter and motor body (see 3.8) - major defect - code no. 16002. A sample of 50 igniter and motor parts assemblies shall be selected from each production lot and tested as specified in 4.5.1.9 using test equipment as specified in 4.1.2. The lot shall be rejected if two or more assemblies are classed defective. If one assembly is classed defective, a second sample of 100 assemblies shall be selected and tested. The lot shall be rejected and further testing discontinued if and when one more assembly is classed defective.

4.4.3.10 Strength of assembly of closure to motor (see 3.11 and drawing 9235663) - major defect - code no. 08005. A sample of 50 complete rockets shall be selected from each production lot and subjected to the test specified in 4.5.1.10 using the test equipment specified in 4.1.2. The lot shall be rejected if one or more assemblies are classed defective.

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4.4.3.11 Rocket weight (see 3.12 and drawing 9271631) - major defect - code no. 10016. This test shall be performed 100 percent on all lots. This test shall be performed as specified in 4.5.1.11 using test equipment specified in 4.1.2. Defective assemblies shall be rejected and removed from the lot.

4.4.4 Proving ground tests.

4.4.4.1 Test to be performed. For all lots, a sample of fifty weapons shall be randomly selected and tested for compliance with requirements of section 3. A listing of tests to be performed and the quantities alloted to each type of test are indicated in Table II. It is required that the 70°F accuracy and penetration tests be performed prior to the 140°F firing phase in the event replacement weapons are needed to fulfill the total quantities needed for these tests. Every fifth (5th) lot, ten (10) samples shall be selected from the 70° accuracy phase tests and used for the penetration failure requirement tests. If penetration tests fails, continue penetration tests next two (2) consecutive lots.

4.4.4.2 Classification of defects. Defect classifications and code numbers for the proving ground portion of first article testing and production lot acceptance testing shall be as follows:

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Premature functioning of the fuze, warhead, or rocket motor	Visual	17001
2.	Motor rupture or burst prior to impact	Visual	17002
3.	Closure rupture or burst prior to impact	Visual	17003
4.	Ignition delay from the striking of the primer by the firing pin until the emergency of the rocket from the launcher (T_I) in excess of three seconds	Measured	17004
5.	Launcher explodes, implodes, or vents injurious gases or particles through sidewalls	Visual	17005
6.	Fuze becomes armed during drop or TV test	Radiographic	17014
Major:			
101.	Warhead fails to function electrically upon target impact	Visual	17006
102.	Accuracy failure at +70°F	Measured	17007
103.	Misfire	Observed	17008
104.	Mechanical firing pin delay (T_L) from the trigger mechanism until the firing pin strikes the primer in excess of one second-	Measured	17009

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Categories	Defects	Method of Inspection	Code No.
105.	Ignition delay from the striking of the primer by the firing pin until the emergence of the rocket from the launcher (T_I) in excess of one second but not in excess of three seconds	Measured	17010
106.	Parts missing or damaged	Visual	17011
107.	Rocket fails to impact target	Visual	17012
108.	Warhead fails to penetrate target	Visual	17013
109.	Total warhead dud	Visual	17015 (17006*)
110.	Fuze fails to function	Visual	17016

*Production sample, Table II

4.4.4.3 Rejection criteria. With the exception of accuracy criteria, the rejection criteria for the proving ground portion of lot acceptance testing shall be in accordance with table II. The lot shall also be rejected if, during the 70°F accuracy phase, the displacement of the average center of impacts from the center of the target, plus or minus the product of 1.55 and the sample standard deviation exceeds 45 inches vertical distance from target center. Sample standard deviation shall be computed using the formula in 6.4. For defect code 17013, one partial penetration shall be disregarded providing that bulging or cracking at the back side of the plate occurs. Any misfire during penetration tests shall count against the total misfire criteria but shall not count as a warhead or fuze failure. Any fuze failure during penetration tests shall count against the total fuze failure criteria but shall not count as a penetration failure. Weapons which misfire during penetration tests and rounds suffering fuze failure during penetration tests shall be replaced with samples from the 140°F flight tests until the total required quantity of these specific functions has been tested. High temperature tests (140°F) shall be conducted using balance of samples available for that test.

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TABLE II

PRODUCTION SAMPLE

ACCEPTANCE/REJECTION CRITERIA PROVING GROUND TESTS

DEFECT Code	QUANTITY IN TEST / TEST / DEFECT	10 -40°F FUNCTION	10 +140°F FUNCTION	10 WATERPROOF PENETRATION	20 +70°F ACCURACY	QUANTITY CONSIDERED	CRITERIA ACC.	CRITERIA REJ.
17001	Premature	X	X	X	X	50	0	1
17002	Motor Rupture	X	X	X	X	50	0	1
17003	Closure Rupture	X	X	X	X	50	0	1
17004	T_I > 3 Seconds	X	X	X	X	50	0	1
17005	Launcher Fails	X	X	X	X	50	0	1
17006	Warhead Duds and Elec Failure	X	X	X	X	50	2	3
17007	Accuracy			X*	X	20-30	*	*
17008	Misfire	X	X	X	X	50	1	2
17009	*** T_L 1 to 3 Seconds	X	X	X	X	50	1	2
17010	**** T_I 1 to 3 Seconds	X	X	X	X	50	1	2
17011	Parts Missing or damaged	X	X	X	X	50	0	1
17012	Target Miss	X	X	X	X	50	1	2
17013	Penetration Failure			X**		10	0	1
	Test Method	4.5.2.1.3	4.5.2.1.3	4.5.2.1.1	4.5.2.1.2			
	Requirement	3.15.3	3.15.4	3.15.2	3.14.3			

* See 4.4.4.3
** See 4.4.4.1
*** Time to Launch (see 6.7)
**** Time to Ignition (see 6.8)

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4.5 Test methods and procedures.

4.5.1 Loading plant tests.

4.5.1.1 Physical strength of exit cone test. The motor assembly shall be assembled to the test equipment specified and the required stresses applied. Motor bodies that fracture, undergo permanent deformation, or burst shall be classed defective. The test equipment shall be functionally verified at four hour intervals of production and immediately prior to the beginning of and immediately following the completion of acceptance operations for the day. This verification shall consist of stressing standards capable of determining whether stress application to the motor body is within the specified limits and gaging to determine the internal hydrostatic pressure. Stress application measurements shall be centered at the specified points. All test equipment verification shall be witnessed by the Government Inspector who may conduct verification at any unscheduled period or periods of production. The Government Inspector shall reject all motor bodies accepted since the last successful verification subject to retest by the contractor witnessed by the Government Inspector, if during any verification the test equipment fails to be applying the minimum force required as determined by the test equipment standard and the hydrostatic pressure measurement. The Government Inspector shall reject all motor bodies accepted since the last successful verification, if during any verification the test equipment is determined to be applying force resulting in stress in excess of the maximum specified, as determined by the test equipment standard and the hydrostatic pressure measurement. This test shall be accomplished without applying metal-to-metal contact between test equipment and the nozzle cone or throat surface in the test specimen.

4.5.1.2 Physical defects tests. The motor body assembly shall be placed on the specified test equipment and the entire inner nozzle exit cone shall be scanned by the probe of the eddy current type test apparatus utilizing the "phase shift" method of detection. Standard bodies with designed defects shall be utilized to calibrate the test equipment for the proper accept or reject signal and proper stamping operation. The motor assembly shall be classed defective and removed from the lot if after having been subjected to the test the proper acceptance mark is not present on the motor.

4.5.1.3 Test for cracks in the closure. Closures shall be placed on the specified test equipment and the entire pressure cavity wall and base including the threads shall be scanned using the applicable inspection equipment. Standard closures with designed defects shall be utilized to calibrate the equipment for consistent reproduction of a signal of the magnitude specified in the operating and calibration procedure. The inspection equipment gate shall be adjusted to the level specified in the operating and calibration procedure. The test standard closures shall then be tested for a consistent

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rejection signal. Acceptable closures only, shall be stamped with proper marking indicating the closure has been subjected to the test and was not rejected. Closures containing defects which produce signal indications equal to or in excess of the gate level specified in the operation and calibration procedures shall be classed defective and removed from the lot. All closures stamped since the last successful equipment calibration verification shall be rejected, subject to retest by the contractor witnessed by the Government Inspector, if during any verification, the test equipment fails to reject the test standards or if the test standards fail to produce a signal of the specified magnitude or if the gate level fails to be operating at the specified level.

4.5.1.4 Continuity of cable assembly in loaded warhead without fuze. The assembly shall be assembled in the test fixture specified in 4.1.2 and each conductor circuit resistance measured. Any assembly that fails to comply with the specified requirement shall be classified defective and removed from the lot.

WARNING: When measuring the resistance the test equipment shall limit the current to 20 microamperes to prevent inadvertant functioning of the detonators. This test may be dangerous and shall be performed behind a suitable barricade.

4.5.1.5 Electrical resistance of warhead assembly. The electrical resistance shall be measured between the positive and negative contacts of the nose up connector on the assembly consisting of loaded warhead, fuze, and closure after bonding. The assembly shall be classed defective if the resistance is less than the specified requirement.

WARNING: When measuring the resistance the test equipment shall limit the current to 20 microamperes to prevent inadvertant functioning of the detonators. This test may be dangerous and shall be performed behind a suitable barricade.

4.5.1.6 Closure/warhead joint pull test. After application and cure of the adhesive, the assembly shall be subjected to a longitudinal static load of 200 lbs $\pm$ 10 lbs for a period not less than 5 seconds. Acceptable assemblies shall be stamped with the proper marking indicating the assemblies have been subjected to the test and were not rejected. Any assembly which separates or becomes loose shall be classed defective.

4.5.1.7 Cure strength of unassembled coupler. The length of cured, impregnated fiberglass shall be cantilevered as specified in 3.9. The deflection of the cantilever shall then be measured. Any test specimen with deflection greater than the requirement shall be classed defective.

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4.5.1.8 Moisture content of black powder. The black powder shall be emptied from the igniter body into a tared weighing dish, accurately weighed, and dried for 4 hours at 70 to 75 degrees centrigrade. The dish shall be covered, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of moisture in the black powder.

4.5.1.9 Seal between igniter and motor body. The motor body shall be conditioned between 65 degrees F and 75 degrees F for a minimum period of three hours prior to performing this test. The test shall be completed within one minute after removal from the conditioning chamber. The motor body with the igniter assembly shall be assembled to the required test equipment. The exit cone interior of the motor body shall be sealed off by the test equipment and a controlled pressure of 10 pounds per square inch minimum of clean dry air applied to the propellant chamber of the motor body. Any igniter-motor body throat seal which permits a leakage rate equal to or in excess of the rejection test standard shall be classed defective. Leakage rate shall be determined by a reading of flow rate or pressure increase taken in the exit cone interior of the motor body. Test equipment sensitivity shall be verified prior to acceptance operations and at one hour maximum intervals of testing. Test equipment verification shall consist of testing both an acceptance and a rejection test standard over a common time interval which will be maintained throughout testing, and observing for proper acceptance or rejection signal. The test standards shall consist of motor bodies with throat seals which permit no leakage other than that resulting from a hole of a specified size located at a common location in the throat seal of each test standard. The rejection test standard shall contain a .002 maximum diameter hole and the acceptance test standard shall contain a smaller hole.

4.5.1.10 Torque strength of rocket motor and closure assembly. The assembled rocket shall be subjected to the specified disassembly torque when the torque is applied between the motor body and the closure. The assembly shall be classed defective if movement between the motor body and closure is observed.

4.5.1.11 Rocket weight. Rockets shall be weighed and light rounds may be corrected by wrapping lead tape about the closure. Rounds which exceed the weight requirement shall be rejected and removed from the lot.

4.5.2 Proving ground test. For all 200-meter range flight tests, a 7-1/2 x 7-1/2 foot target shall be inscribed on a 10 x 10 foot surface placed perpendicular to the line of sight to the firing point surveyed and positioned at 200 meters. Rounds shall be conditioned at the required temperature for a minimum of 18 hours and then shall be secured in the accuracy test holding fixture and fired within three minutes after removal from conditioning. If the rocket has not been fired within three minutes, the round shall be conditioned at the required temperature for one hour for each additional minute, but in no case will more than 18 hours be required. Individual aiming of each complete round shall be accomplished at the angle of elevation specified as follows:

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Temperature

64.6 mils for the -40°F
53.4 mils for the +70°F
49.1 mils for the +140°F

With the accuracy test sighting fixture positioned as specified above and the launcher secured in the flight test holding fixture, the holding fixture position shall be adjusted until the telescopic sight of the accuracy test sighting fixture is aimed at a point on the accuracy test target as specified in 4.1.2. Sighting procedure for accuracy test of M72A2 weapon shall be as follows:

a. Sighting fixture shall be placed in an empty M72A1 launcher and the proper QE for the test temperature shall be established.

b. Panoramic telescope of firing fixture shall be adjusted to center of target.

c. Empty M72A1 launcher shall be replaced with M72A2 weapon to be tested.

d. Panoramic telescope shall be checked prior to firing each round to insure that the firing fixture has not been moved. If the firing fixture has moved, paragraph a, above, shall be repeated.

A steel launcher support fixture with an azimuth and quadrant elevation adjustment mechanism, shall be mounted on concrete poured below freezing line in the ground so that the entire fixture will be rigid for about 12 lb-sec-load from the launcher.

Flight testing of the weapon at the 200 meter target shall not be conducted with wind conditions in excess of 10 knots. Upon firing, time elapsed shall be recorded between when the trigger mechanism is activated and the firing pin strikes the primer and between when the firing pin strikes the primer and the emergence of the rocket from the launcher. To distinguish between electrical and graze functioning on target impact, the following characteristics are generally reliable. Electrical functioning is evidenced by a small diameter penetration while a large diameter penetration indicated graze functioning and is generally accompanied by an indented ring around penetration. The coordinates of impact shall be measured vertically from the lower edge and horizontally from the left side of the 10' x 10' target, and all measurements shall be in inches to the nearest 1/2 inch. Velocities and impact coordinates shall be recorded in all records and reports along with the ambient temperature, wind direction and wind velocity.

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Elevation angle note: Angle of elevation is between the centerline of the telescopic line of sight of the accuracy test sighting fixture and the centerline of the launcher bore required for each of the conditioning temperatures during accuracy and functioning tests.

4.5.2.1 Production testing.

4.5.2.1.1 Waterproofness and penetration. The weapon shall be placed horizontally under three feet of water for two hours at 70°F ± 5°F. After having been submerged for the required time, the weapon shall be removed from the water, the sling assembly removed and the launcher with the rocket shall be held vertically for approximately five (5) seconds, then turned so that the upper end is down for five (5) seconds. The weapon shall then be aimed at the penetration test target and functioned. Performance shall be measured against each of the applicable defects listed in 4.4.4 and Table II, and results recorded.

4.5.2.1.2 Accuracy and functioning at 70°F. This test shall be conducted prior to the 140°F tests. Prior to firing of this test, the rockets shall be inspected for parts missing or damaged. (Defect Code 17011). If any parts are missing or damaged which in the opinion of the testing agency would affect safe functioning or accuracy, said rocket shall be removed from the test and replaced with rockets selected from the 140°F test phase. All weapons shall be conditioned and fired as indicated in 4.5.2. Performance shall be measured or observed and evaluated against each of the applicable defects listed in 4.4.4 and Table II.

4.5.2.1.3 Functioning at -40°F and +140°F. Rounds shall be conditioned and fired as indicated in 4.5.2. Performance shall be measured or observed and evaluated against each of the applicable defects listed in 4.4.4 and Table II, and results recorded. Impact coordinates shall be reported for information purpose only.

4.5.2.2 Test peculiar to first article.

4.5.2.2.1 Transportation-vibration test. The weapon shall be placed in cartons conforming to dwg. 9227925 and tested in accordance with MIL-STD-810, Method 514.1, Procedure X, Curve AA. Performance shall be evaluated against each of the applicable defects listed in 4.4.4 and Table I and results recorded.

WARNING: After Transportation-Vibration Test and before further conditioning or test, the test items must be radiographically examined in-place and without being moved to assure that fuzes are not armed.

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After test and examination the weapon shall be visually examined for damage and any evidence of damage recorded in the test record. Any condition which would render the weapon unsafe to move or fire must be reported immediately. Items thus tested which appear to be serviceable shall be conditioned and tested in accordance with 4.5.2.1.3 (140°F only) and results reported. Unserviceable items shall be disposed of.

4.5.2.2.2 Drop tests. The closed weapon in carrying configuration shall be suspended from a height of five (5) feet to its lowest most point and dropped remotely onto a concrete surface from the horizontal orientation allowing impact to occur on the side opposite the trigger housing.

WARNING: After drop test, the item must be radiographically examined on site before being moved by hand and assurance obtained that the fuze is not armed.

After test and examination the weapon shall be visually examined for damage and any evidence of damage recorded in the test record. Any condition which would render the weapon unsafe to move or fire must be reported immediately. Items thus tested which appear to be serviceable shall be conditioned and tested in accordance with 4.5.2.1.3 (140°F only) and results reported. Unservicable items shall be disposed of.

4.5.2.2.3 Graze functioning. The weapon shall be conditioned and fired at -40°F in accordance with 4.5.2 at an impact area of hard packed earth at approximately 200 meters range (minimum). Performance shall be measured against each of the applicable defects listed in 4.4.4 and Table I and results recorded.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The complete rounds shall be preserved and packaged in accordance with 9227925.

5.2 Packing. The complete rounds shall be packed in accordance with dwg. 9227926.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

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6.2 Inspection code numbers. The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These codes numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.3).

6.3. Inspection equipment designs. Inspection equipment designs are of two types - Government special inspection equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the Method of Inspection headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Drawings shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit code number (or other specific identifying information) from

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Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-R, Dover, New Jersey 07801, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Calculation of sample standard deviation. The sample standard deviation will be calculated using the following formula or its equivalent:

$$s = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n - 1}}$$

$\bar{x}$ = Average value for the sample
x = Numerical value for individual observations
s = Sample standard deviation
n = Sample size

6.5 Note on ignition delay or hangfires. It is possible to separate explosive and mechanical delays by means of two chronographs, one to be started by the firing solenoid impulse and stopped by the primer function acoustically, and a second started by primer function and stopped as the rocket enters the first velocity coil.

6.6 Submission of first article sample. Instructions as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receive of each request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-I
Dover, New Jersey 07801

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6.7 Launcher reaction time (T_L) is defined as the delay between activation of the trigger mechanism until the firing pin strikes the primer.

6.8 Ignition time (T_I) is defined as delay between the striking of the primer by the firing pin until emergence of the rocket from the launcher.

6.9 The margins of this specification are not marked to denote changes from the previous issue due to the extensiveness of the changes incorporated.

Custodian:
Army - MI

Review Activity:
Army - PA

Preparing Activity:
Army - MI

Project No. 1340-A414

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Commander
US Army Missile Command
ATTN: AMSMI-RCS
Redstone Arsenal, AL 35809

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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